

Lab - Use Steganography to Hide Data

Objectives

Use steganography to hide a document within a JPEG file.

Background / Scenario

The advantage of steganography over cryptography is that the secret message does not attract any special attention. No one would ever know that a picture contained a secret message by viewing the file either electronically or in hardcopy. In this lab, you will use Steghide, an open-source steganography program, to hide a data file within an image file.

Required Resources

PC with the **CSE-LABVM** installed in VirtualBox

Instructions

Step 1: Open a terminal window in the CSE-LABVM.

- Launch the **CSE-LABVM**.
- Double-click the **Terminal** icon to open a terminal.

Step 2: Review the files that will be used for steganography.

- Enter the **cd Downloads/** command to change to the **Downloads** directory, and then list the contents of the directory.

```
cisco@labvm:~$ cd Downloads/
cisco@labvm:~/Downloads$ ls -l
total 472
drwxr-xr-x 8 cisco cisco 4096 Jan 27 01:30 jcryptool
drwxrwxr-x 3 cisco cisco 4096 Mar 18 18:11 john
-rw-rw-rw- 1 cisco cisco 455357 Mar 22 16:10 keyboard.jpg
drwxrwxr-x 6 root root 4096 Mar 18 17:46 lynis
-rw-rw-r-- 1 cisco cisco 9912 Mar 22 16:13 secret.odt
cisco@labvm:~/Downloads$
```

You will hide the contents of the secret.odt file inside the keyboard.jpg file.

- Enter the **libreoffice secret.odt &** command to open the "secret.odt" file in LibreOffice.

```
cisco@labvm:~/Downloads$ libreoffice secret.odt &
```

What is the message in the **secret.odt**?

The secret document

- Click **File > Exit LibreOffice** to quit LibreOffice and close the file.
- In the terminal window, press **Enter** to get a new command prompt, and then enter command **gimp keyboard.jpg &** to open the "keyboard.jpg" file in GIMP.

```
cisco@labvm:~/Downloads$ gimp keyboard.jpg &
```

- e. Click **File > Quit** to quit GIMP and close the file.
- f. In the terminal window, press **Enter** to get a new command prompt.

Step 3: Use Steghide to embed the content of the secret.odt file inside the keyboard.jpg file.

- a. At the command prompt, enter the following **steghide** command. When prompted for a passphrase, use **Cisco**. Re-enter the passphrase when prompted.

```
cisco@labvm:~/Downloads$ steghide embed -cf keyboard.jpg -ef secret.odt
Enter passphrase: Cisco
Re-Enter passphrase: Cisco
embedding "secret.odt" in "keyboard.jpg"... done
cisco@labvm:~/Downloads$
```

- b. Open the files, **secret.odt** and **keyboard.jpg**.

Did these files change?

No change was found

Step 4: Verify the secret.odt is hidden in the keyboard.jpg file.

Enter the **steghide info keyboard.jpg** command, type **y** at the prompt, and enter the passphrase **Cisco**, and then press **Enter**.

```
cisco@labvm:~/Downloads$ steghide info keyboard.jpg
"keyboard.jpg":
format: jpeg
capacity: 26.6 KB
Try to get information about embedded data ? (y/n) y
Enter passphrase: Cisco
embedded file "secret.odt":
size: 9.7 KB
encrypted: rijndael-128, cbc
compressed: yes
cisco@labvm:~/Downloads$
```

Step 5: Extract the secret.odt file from the keyboard.jpg file.

- a. Enter the **steghide extract -sf keyboard.jpg** command, enter the passphrase Cisco, and then type **y** at the prompt.

```
cisco@labvm:~/Downloads$ steghide extract -sf keyboard.jpg
Enter passphrase: Cisco
the file "secret.odt" does already exist. overwrite ? (y/n) y
wrote extracted data to "secret.odt".
cisco@labvm:~/Downloads$
```

- b. Open the extracted **secret.odt** file with LibreOffice.

Could you open the file? Is the secret message the same as before?

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Yes and Yes

```
cisco@labvm: ~/Downloads
File Edit View Search Terminal Help
cisco@labvm:~/Downloads$ cd Downloads/
cisco@labvm:~/Downloads$ ls -l
total 604
drwxr-xr-x 8 cisco cisco 4096 Jan 27 2021 jcryptool
drwxrwxr-x 3 cisco cisco 4096 Mar 18 2021 john
-rw-rw-r-- 1 cisco cisco 409604 Mar 18 2021 keyboard.jpg
drwxrwxr-x 6 root root 4096 Mar 18 2021 lynIs
-rw-rw-r-- 1 cisco cisco 179886 Apr 23 18:13 'Screenshot at 2026-04-23 18-13-27.png'
-rw-rw-r-- 1 cisco cisco 9912 Mar 18 2021 secret.odt
cisco@labvm:~/Downloads$ libreoffice secret.odt &
[1] 11461
cisco@labvm:~/Downloads$ Warning: failed to launch jvaldx - java may not function correctly
[1]+ Done libreoffice secret.odt
cisco@labvm:~/Downloads$ gimp keyboard.jpg &
[1] 11520
cisco@labvm:~/Downloads$ Gtk-Message: 17:55:56.689: Failed to load module "gall"
** (gimp:11520): WARNING **: 17:55:56.787: (./atk-adaptor/bridge.c:1018):atk_bridge_adaptor_init: runtime check failed: (root)
Gtk-Message: 17:56:05.103: Failed to load module "gall"
** (file-jpeg:11675): WARNING **: 17:56:05.205: (./atk-adaptor/bridge.c:1018):atk_bridge_adaptor_init: runtime check failed: (root)
[1]+ Done gimp keyboard.jpg
cisco@labvm:~/Downloads$ steghide embed -cf keyboard.jpg -ef secret.odt
Enter passphrase:
Re-Enter passphrase:
embedding "secret.odt" in "keyboard.jpg"... done
cisco@labvm:~/Downloads$ steghide info keyboard.jpg
"keyboard.jpg":
  format: jpeg
  capacity: 26.6 KB
Try to get information about embedded data ? (y/n) y
Enter passphrase:
  embedded file "secret.odt":
    size: 9.7 KB
    encrypted: rijndael-128, cbc
    compressed: yes
cisco@labvm:~/Downloads$ steghide extract -sf keyboard.jpg
Enter passphrase:
the file "secret.odt" does already exist. overwrite ? (y/n) y
wrote extracted data to "secret.odt".
cisco@labvm:~/Downloads$
```